

Problem Identification & Project Specification

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ABSTRACT

Predicting whether the price of a financial asset will rise or fall over a given period is an important problem for investors and traders. Machine learning models can analyze large amounts of historical market data to identify patterns that traditional econometric models, which rely on linear assumptions, are unable to capture. Common approaches include tree-based models such as Random Forest and AdaBoost, support vector machines, and deep learning models such as LSTM and GRU networks, which have been applied to both stock and cryptocurrency markets. Handling noisy and non-stationary data, selecting informative features, and maintaining interpretability are key challenges for improving model performance. This project aims to build a binary classification model using publicly available cryptocurrency and stock market datasets, applying established machine learning techniques to determine whether an asset will increase or decrease in price after a specified time window.

KEYWORDS

Market trend prediction, Machine learning, Binary classification, Cryptocurrency

1. Introduction

The stock market is one of the cornerstones of economies worldwide. It is a marketplace where investors buy and sell shares of companies that are listed publicly. The share prices fluctuate based on supply and demand and are influenced by economic factors, corporate performance, news, and global events [1]. The stock market can be studied as it is based on rules and mechanisms, has data and models that involve statistics and economics, and has theories and hypotheses that explain market trends. Assessments can be made by looking at the value of the stock itself, but also by analysing the market activity. But at the same time, prices are affected by human factors like investor optimism or pessimism, fear, and greed. And there are no guarantees that the rules applied

and the phenomena observed today will hold later as human behaviour changes [5]. So even though the stock market can be understood using the methods discussed above, predicting whether a stock will rise or fall after a certain period of time remains a complex task that requires specialized studies, analysis, and sometimes trial and error.

In recent years, the new market of cryptocurrencies has emerged and is becoming increasingly significant for investments and trading, and important efforts are being made to scientifically understand and analyse the market and its dynamics. Cryptocurrency returns are not as symmetric as stocks, are more volatile, and are even more dependent on speculations and investor and the community's mood. It is also important to note that it is a 24/7 continuous market, unlike the stock market that has fixed operation hours [2].

Applying a machine learning approach in such a field is beneficial because of the presence of large amounts of data, which allows us to detect previous patterns and help us predict future trends. We intend to build a binary classification model that takes as input a certain asset and a time window and determines whether, after this designated period of time, the asset will increase or drop in price.

2. Literature Review

Prices of assets like stocks and cryptocurrency can be considered as time series data. Initially, econometric models like AR and Auto Regressive Moving Average (ARMA) were being used to make predictions based on previous data, but these models were found to be lacking because of their linear assumptions and due to the complex nature of the data.

Studies later found that machine learning is able to give better results and therefore many ML approaches have been studied. Some of these approaches are linear models, like Support Vector Regression (SVR), AdaBoost, Random

Forest (RF), and Naïve Bayes algorithm. These models have shown high predictability, even though some might have higher risks of overfitting than others. These models also deliver data that can be interpreted using explainability framework tools like weighted impurity (Gini index), Partial Dependence Plots (PDP) and SHAP (SHapley Additive exPlanations) values. This is extremely valuable as to understand the reasons behind the market's volatility and dynamics. [2, 6]

We eventually also started looking at deep learning (DL) models, like Long Short-Term Memory (LSTM), Gated Recurrent Units (GRU), Multilayer Perceptrons (MLP), Artificial Neural Networks (ANN), and Support Vector Machine (SVM) [1, 2]. Some approaches even go towards Natural Language Processing (NLP) to analyse forums, social media, and other platforms where investors express their opinions, which is very valuable information. These models give better performance in predicting volatility and detecting mood shifts in the market, but they are less interpretable than traditional ML models and require extensive resources and are outside of our scope for this project. [6]

3. Methodology

We will follow a phased approach where each phase has distinct goals and deliverables. This will ensure clear progress tracking and more efficient results being delivered.

3.1 Problem Identification and Project Specification: We are studying the literature and identifying which previous approaches worked or did not and the reasons behind them. We are also identifying datasets and will choose the one to use in order to train our machine learning model.

3.2 Data Preparation, Cleaning, and Feature Generation: We will need to report on our chosen dataset, clean the data, perform pre-processing and feature engineering to obtain the most information out of the data. After doing so, we will need to communicate the steps taken, the reasons behind every decision taken, and describe the generated features.

3.3 Pilot Study on Models: We will perform a study where we will measure and compare the performance of different supervised learning models, specifying the strong and weak points of each one. This evaluation will be based on our selected prepared data and will be made through

conventional performance measures such as the F-score and the AUC-ROC.

3.4 Model Implementation and Application: In this phase, we will be implementing the final decided model and will be training, testing, and tuning it until getting the best performance. And we will create an application showcasing our model where users can select a certain stock and a time window and get as output whether the stock will rise or fall, depending on the model's classification.

4. Dataset Description

Binance 1-Minute Klines [8] This dataset contains one-minute candlestick records for cryptocurrency trading pairs collected from the Binance public market data API, covering price and trading activity for each minute of trading. The dataset is available on HuggingFace. It contains approximately 9,001,592 instances described by 12 features, and is around 425 MB in size. The features include the opening, high, low and closing prices for each minute, the traded volume, the quote asset volume, the number of trades executed, and the taker buy base and quote asset volumes, which record the portion of the volume that was initiated by buyers. Unlike churn datasets, this data does not include a target variable, so the label will be constructed by comparing the closing price of each minute with the closing price a fixed number of minutes later, labelling the instance as 1 if the price increased and 0 otherwise, which is the standard approach used in previous studies of directional price prediction [5].

This dataset is highly relevant to our project since it provides information that most price datasets do not, particularly the taker buy volumes, which allow us to calculate the proportion of trading that was buyer initiated and use it as a predictive feature. The number of trades feature also allows us to estimate the average size of each trade, which helps distinguish periods dominated by a few large orders from periods of many small ones. Moreover, these features are identical to those returned by the live Binance API, which will simplify the real-time data collection and retraining component planned for Phase 4.

Despite these strengths, there are some limitations. The dataset is a static archive and is not updated, so live data will need to be collected separately through the API for the final phase. Only trading pairs that are currently listed on the exchange are included, meaning that pairs delisted during

the period are absent and the sample is biased towards surviving pairs. The dataset also does not specify a license, which limits how the data may be redistributed.

Dataset II: High-Frequency Crypto Limit Order Book

[3] This dataset contains limit order book snapshots for Bitcoin, Ethereum and Cardano, reconstructed from raw order book messages collected from the Coinbase exchange, and is available on Kaggle. It covers approximately twelve days of trading and contains around 3.1 million instances at one-second resolution, described by roughly 122 features. The features describe the fifteen best price levels on each side of the order book and include the midpoint price, the spread between the best bid and ask, the percentage distance of each level from the midpoint, and the volume of market orders, limit orders and cancelled orders at each level. As with Dataset I, no target variable is provided, and the label would be constructed from the movement of the midpoint price.

This dataset is relevant to our project because it contains information about orders that were placed but never executed, which cannot be recovered from price data alone. The cancellation features are particularly valuable, since an order book that appears to contain a large amount of liquidity may not actually provide it if those orders are withdrawn as the price approaches them. Including these features would allow us to examine whether information about resting liquidity improves predictive performance compared with the trading volume features available in Dataset I.

However, the dataset presents several challenges. The time period covered is very short, and twelve days of data reflects only a single set of market conditions, which limits how well any model trained on it would generalize. The data was collected in 2021 and is not updated, so it cannot support the live component of the project. It also originates from Coinbase rather than Binance, so the behavior observed may not transfer directly. Finally, the large number of closely correlated features would require substantial feature selection to avoid overfitting.

Dataset III: FI-2010 Benchmark Dataset [4] The FI-2010 dataset is a benchmark dataset created for the task of predicting mid-price movement from limit order book data, and is available through the Kaggle platform and the Finnish research data portal. It contains ten consecutive days of trading for five stocks listed on the Nasdaq Nordic exchange, comprising 394,337 instances described by 144 features.

The features include ten levels of bid and ask prices and volumes, derived quantities such as spreads and mid-price differences, and measures of how frequently orders arrive. Unlike the other datasets considered, FI-2010 includes a target variable, in which each instance is labelled as an upward, stationary or downward movement of the mid-price across five different prediction horizons.

This dataset is relevant to our project mainly as a point of comparison rather than as a source of training data. Classifiers predicting price direction typically achieve accuracies close to 50%, which makes it difficult to judge whether a given result represents genuine predictive performance without an external reference. Since results for logistic regression, support vector machines and neural networks have been published on this dataset, it would allow us to position our own Phase 3 results against established work. The included labels also provide a worked example of how the prediction horizon and the threshold separating an increase from a flat movement have been chosen in the existing literature.

The dataset does however have clear limitations. Ten days of trading across five anonymized stocks is a small sample and covers a narrow range of market conditions. The labels use three classes rather than two, so we would need either to discard the stationary instances or to adopt a comparable threshold in our own labelling. The stationary class is also considerably larger than the other two at short horizons, which illustrates the class imbalance that our own choice of threshold is likely to produce.

Dataset IV: US Stock Market Daily History [7] This dataset contains daily historical price data for over 9,000 stocks traded on United States exchanges between 1962 and 2024, and is available on Kaggle. It is approximately 4.47GB in size. The dataset comprises seven features, namely the date, the ticker symbol, and the opening, high, low and closing prices together with the traded volume for each day. As with the cryptocurrency datasets, no target variable is included and the label would need to be constructed from subsequent closing prices.

This dataset is included for the purpose of comparison rather than for training our model. One of the requirements of this phase is to contrast the continuous nature of cryptocurrency markets with the fixed trading schedules of conventional exchanges, and this dataset allows that comparison to be made using actual data. Because stock markets close

overnight and at weekends, technical indicators calculated over a rolling window necessarily span periods during which no trading occurred, whereas cryptocurrency markets trade continuously and are not affected by this issue.

The limitations of this dataset are considerable for our purposes. The daily frequency of the data is far too low for the high-frequency task we are addressing, so it cannot be used to train the main model. The prices have been adjusted for stock splits and dividends, which means that historical rows incorporate corporate actions that would not have been known at the time they were recorded. In addition, the dataset only includes companies that were still trading in 2024, and therefore excludes delisted and bankrupt firms, introducing survivorship bias.

Selection Of the four datasets considered, we intend to use Dataset I as the primary source of training data, as it is the only dataset that records which side initiated each trade while also providing a sufficiently large sample and a feature set matching the live API required for the final phase of the project. Dataset II will be used to investigate whether order book information adds predictive value, Dataset III to compare our results against published work, and Dataset IV to support the comparison between continuous and scheduled markets.

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